Bootstrapping Vision-Language-Action Models for New Robot Embodiments with Staged Reinforcement Learning

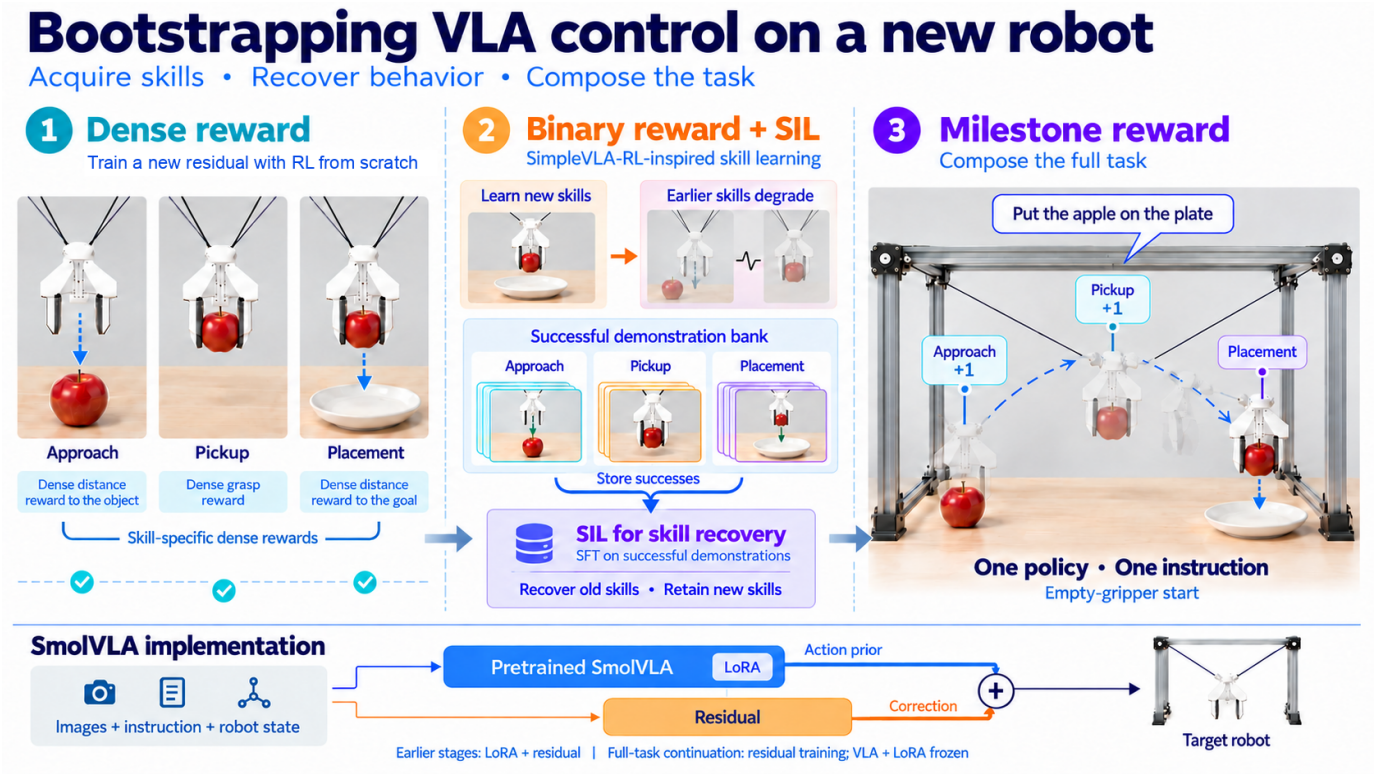


Fig. 1. Proposed pipeline and SmolVLA implementation. Dense feedback acquires skills; binary refinement and balanced self-imitation consolidate them; milestone-based RL learns the complete task under one instruction. Skill-specific resets are used during acquisition. Full-task evaluation starts with an empty gripper and uses no external switching between specialists.

***Abstract*—Adapting a pretrained vision-language-action (VLA) model to a new robot commonly requires demonstrations collected on that embodiment. We propose a staged reinforcement-learning pipeline for adaptation without collecting human demonstrations on the target robot. Dense rewards first guide the acquisition of simple skills. Binary refinement and balanced self-imitation consolidate these skills, followed by full-task learning with binary milestones. The milestones provide intermediate credit without requiring a continuous progress function for every phase of the complete task. We evaluate the pipeline in cable-driven parallel robot (CDPR) simulation using a newly initialized residual controller over a pretrained SmolVLA expert, with LoRA adaptation. The controller achieves 69.14% strict full-task success over 2,048 off-panel development attempts. A complementary OpenVLA-OFT experiment on LIBERO-Spatial examines intermediate feedback under a common training setup. Combined dense and phase feedback reaches 65.3% success versus 61.7% with terminal-only reward after 160 updates.**

***Index Terms*—Robot learning, reinforcement learning, manipulation, vision-language-action models, robot adaptation.**

I. Introduction

Vision-language-action (VLA) models provide useful visual and language representations for robot control, but adaptation to a new embodiment commonly relies on demonstrations collected with the target robot [1], [2]. For custom robots, collecting these data requires a suitable control interface and repeated demonstrations of the desired behavior. We study adaptation that begins with reinforcement learning (RL), without collecting human demonstrations on the target embodiment. The difficulty is that a policy transferred to a new action interface may rarely complete the task, leaving binary success with little guidance for learning.

Consider *put the apple on the plate*. A failed rollout may contain a correct approach and grasp followed by a poor release. Treating the entire attempt as a failure discards this distinction. Training the parts separately helps exploration, but a policy can forget earlier skills while learning later ones. It must then connect the retained skills into a complete behavior.

We address these problems through the pipeline in Fig. 1. Dense rewards are used only at the beginning to acquire a base policy for simple skills. Binary refinement then targets skill completion, while balanced self-imitation retains successful behavior across instructions. Finally, staged RL trains this policy to execute the complete multi-stage task from an empty-

gripper start, using binary milestones throughout full-task training.

We evaluate the adaptation pipeline in cable-driven parallel robot (CDPR) simulation using SmolVLA [1], LoRA, and a newly initialized residual controller. This experiment measures skill retention and complete-task performance on the target action interface. A complementary experiment uses OpenVLA-OFT [2] on LIBERO-Spatial to compare terminal and intermediate feedback under a common training setup. It examines the feedback used to guide multi-stage behavior, while the CDPR experiment evaluates the complete acquisition, consolidation, and composition procedure.

We contribute a training schedule that begins target-robot adaptation with dense-reward skill acquisition, followed by binary refinement with instruction-balanced self-imitation and staged full-task learning under one instruction. We specify milestone credit that retains local credit for achieved approach and pickup stages when later placement fails. Our experiments compare intermediate and terminal feedback, measure skill recovery after sequential training, and report complete-task performance after action-interface adaptation.

II. Related Work

Group-relative RL and dynamic sampling. GRPO estimates policy advantages by comparing returns within groups of sampled trajectories [3]. DAPO introduces dynamic sampling that filters groups with uniform binary outcomes and collects additional samples to maintain an informative training batch [4]. SimpleVLA-RL adapts group-relative optimization to interactive robot rollouts, using dynamic sampling and exploration enhancements to improve task execution and discover behaviors beyond the supervised demonstrations [5]. These methods provide a basis for learning task-specific behavior through interaction. We use DAPO-style group filtering and resampling during skill learning.

Alternating RL and supervised learning. iRe-VLA alternates online RL on a lightweight action head with a frozen vision-language backbone and supervised updates using expert data together with successful online trajectories [6]. This separates exploration from supervised consolidation while allowing the large model to benefit from collected experience. Our self-imitation step fits an instruction-balanced mixture of earlier and current successes to retain skills during sequential training.

Residual RL and distillation. Probe, Learn, Distill (PLD) trains lightweight residual specialists over a frozen VLA, collects trajectories through policy-aligned residual interventions, and distills the resulting data into the generalist [7]. We adopt the residual-learning principle to adapt the target action interface. Our training procedure links skill acquisition and successful-experience consolidation before learning to execute the complete instruction.

Our earlier work studied RL bootstrapping of OpenVLA-OFT on the CDPR for directional motion and object reaching [8]. The present work extends the study to complete pick-and-place through skill consolidation and milestone-based learning with a SmolVLA residual controller.

III. Methods

The goal is to adapt a pretrained VLA to complete language-conditioned manipulation tasks on a target robot without collecting human demonstrations. We study multi-stage pick-and-place tasks: an instruction such as *put the apple on the plate* requires approach, grasp, lift, transport, and release. The policy receives camera images, proprioception, and an instruction, predicts an action chunk, and executes a prefix before observing again. We evaluate complete-task success independently of the reward used for training.

During acquisition, each skill has its own instruction, success condition, and starting states. For example, placement can begin with an object already held. During composition and final evaluation, the robot starts with an empty gripper and receives one complete-task instruction. This distinction prevents isolated skill performance from being treated as full-task success.

A. Dense Acquisition, Binary Refinement, and Self-Imitation

We use dense rewards only at the beginning of adaptation to learn a base policy for simple skills. Each skill has a distinct instruction and reset distribution:

- **Approach:** *move to <object>*. Reach the named object with an empty gripper, without requiring a grasp.
- **Pickup:** *pick up <object>*. Learn to grasp and lift the named object.
- **Placement:** *put <object> into bowl* or *put <object> on the plate*. Start with the object already grasped and learn transport and release at the named destination.

The object placeholder is replaced by its name. These instructions condition the policy on the requested skill and target; the instruction remains attached to each retained trajectory. Distance reduction and manipulation events guide initial acquisition. We then refine the skills with binary success rewards and consolidate them through self-imitation to prepare a shared base policy for full-task learning.

Successful trajectories are retained by instruction. When later training degrades earlier skills, self-imitation fits a balanced mixture of earlier and current successes:

$$\mathcal{L}_{\mathrm{SI}} = \mathbb{E}_{\mathrm{success}} \left[\|\hat{a} - a\|^2 \right]. \tag{1}$$

Here $\hat{a}$ is the predicted action and a is the recorded executed action. The average balances instructions and excludes unused action slots. Policy-dependent features are refreshed before fitting. Balancing matters because imitating only the newest skill can reinforce forgetting.

Algorithm 1 summarizes the schedule. Progress is checked with fixed skill evaluations. A stage ends at a success target, a plateau, or budget exhaustion. In the CDPR experiment, transitions were selected from validation performance. After consolidation, all learned skills are evaluated again before proceeding.

B. SmolVLA with an Embodiment-Specific Residual

Figure 2 shows one implementation of the pipeline. Following the residual-learning ingredient of Probe, Learn, Distill

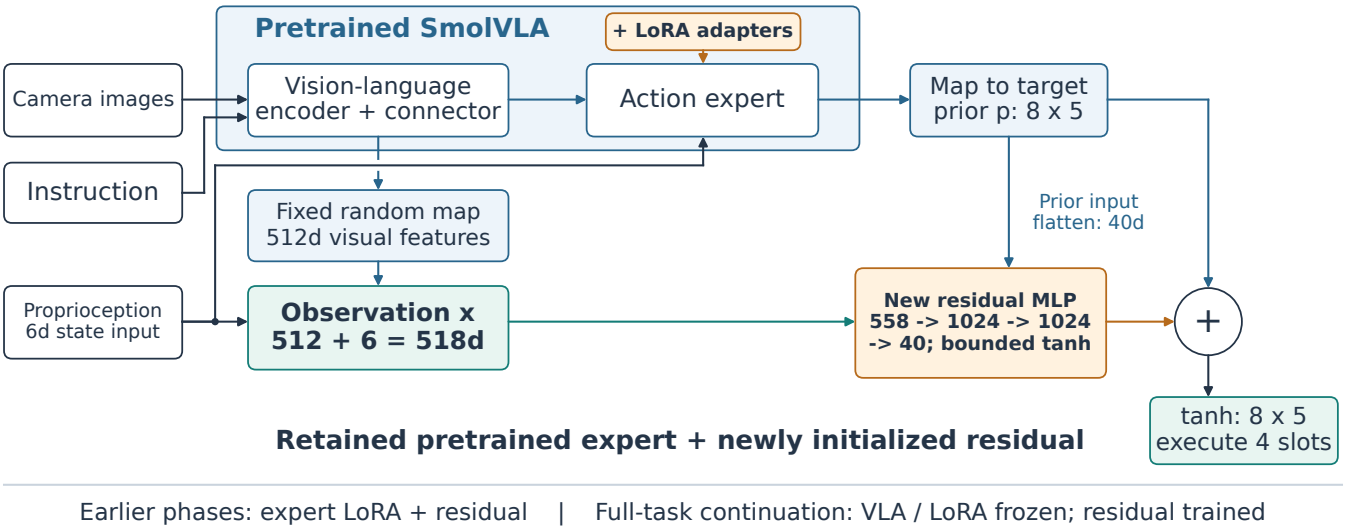


Fig. 2. SmolVLA implementation of the pipeline. The added residual combines 518 observation features with a 40-dimensional action prior. LoRA adapts the pretrained expert during earlier training; the evaluated full-task continuation freezes the VLA and LoRA weights. Feature and action dimensions refer to the five-channel controller.

Algorithm 1 Staged VLA adaptation

Require: Pretrained VLA, simple-skill instructions, fixed evaluations
Load the pretrained VLA; initialize a residual and success banks
for reward mode: dense acquisition, then binary refinement **do**
  for skill: approach, pickup, placement **do**
    repeat
      Collect same-scene rollout groups for the skill
      Update with $\mathcal{L}_{\mathrm{RL}}$; retain successful trajectories
      Evaluate skill success
    until target, plateau, or stage budget ends training
    Evaluate retention of all learned skills
    if earlier skills have degraded **then**
      Balance earlier and current successful experience
      Fit $\mathcal{L}_{\mathrm{SI}}$; validate recovery and retention
    end if
  end for
end for
Initialize full-task learning from the consolidated base policy
repeat
  Collect full-task rollouts from empty-gripper starts
  Assign milestone credit; protect achieved prefix skills
  Update with $\mathcal{L}_{\mathrm{RL}}$; evaluate complete-task success
until target, plateau, or full-task budget ends training
if composition plateaus and budget remains **then**
  Freeze VLA/LoRA; retain the bounded residual
  Reset optimizer; keep milestone rewards unchanged
  repeat
    Collect full-task groups; execute clipped commands
    Store pre-clipping samples and episode offsets
    Score samples conditional on their stored offsets
    Assign milestone credit; protect achieved prefix skills
    Update residual and exploration with $\mathcal{L}_{\mathrm{RL}}$
    Evaluate complete-task success
  until target, renewed plateau, or budget exhaustion
end if
return One policy for the complete instruction

(PLD) [7], we retain the pretrained SmolVLA expert and initialize a new residual controller. The expert supplies an action prior, mapped to the target control interface. In our five-channel controller, eight action slots form a 40-dimensional prior. A fixed random projection of visual tokens supplies 512 features; six proprioceptive values complete the 518-dimensional observation. The residual receives this observation and the 40-dimensional prior, and predicts a bounded correction. The combined action is bounded before execution.

LoRA adapts the action expert during earlier training. In the evaluated full-task continuation, the inherited VLA and LoRA weights are frozen while the residual and exploration parameters are trained. The backbone, action mapping, and feature dimensions can change with the embodiment; the training schedule remains the same.

C. Milestone-Based Full-Task Composition

After skill consolidation, the base policy learns the complete task from empty-gripper starts using binary milestone rewards. The full-task prompt is *put <object> into <destination>*, where the destination is plate or bowl. The full-task instruction stays fixed throughout approach, pickup, and placement. Each milestone marks completion of part of this sequence. Continuous dense shaping is confined to the initial simple-skill acquisition in Section III-A.

In the adapted CDPR controller, approach, held pickup, and final placement serve as milestones for assigning credit within a rollout. The policy executes the complete instruction without an external controller switching between specialists.

The main rule is simple: a later placement failure should not remove the local credit for a successful approach or pickup. We compare milestone outcomes within groups of rollouts from the same scene. Actions receive credit for the milestone they contribute to, while full-task success provides an additional preference for completing the chain. A negative completion bonus is suppressed on already achieved approach

and pickup milestones. Placement receives its completion credit once.

The rule retains the assigned credit for completed milestones while training the remaining task.

The motivation follows from a simple group calculation. For G independent rollouts from one scene, each with success probability p, the probability of observing both success and failure is

$$P_{\text{mixed}} = 1 - p^G - (1-p)^G. \tag{2}$$

With eight rollouts, this probability is only 21.6% at 3% success, but 94.2% at 30% success. Milestones can therefore provide informative comparisons before complete-task success becomes common.

To examine intermediate feedback on a standard benchmark, the LIBERO experiment compares terminal-only, phase-only, and combined dense-and-phase rewards. The combined reward uses a manually defined sequence of reach, grasp, lift, move, and place phases, with a hand-specified progress function for each phase.

At each simulator step, the tracker uses object and gripper state to compute progress for the active phase. Reaching progress increases as gripper–object distance decreases; grasping combines proximity with gripper closure and grasp state; lifting uses height gained above the initial object position; transport uses planar object–target distance; placement uses target proximity, release, and environment success. Each progress measure is normalized to the interval from zero to one. For progress p_t and the phase's previous maximum p_{best}, the rewarded increment is $\max(p_t - p_{\text{best}}, 0)$. A completion predicate advances the tracker to the next phase and produces a phase bonus. The weighted progress increment and phase bonus are bounded and combined with terminal feedback. The native task-success verifier remains unchanged.

In the phase-only condition, the continuous progress weight is zero, leaving phase-completion bonuses and terminal feedback.

The dense construction requires a progress function, normalization, and relevant state variables for every phase. Extending it to a new multi-stage task therefore requires further task-specific reward design. This motivates our use of dense feedback for initial skill acquisition and binary milestones for composition: the latter require completion predicates without a continuous progress function for every part of the instruction.

D. Stabilized Updates and Full-Task Continuation

We use a clipped group-relative policy loss with entropy regularization:

$$\mathcal{L}_{\text{RL}} = \mathcal{L}_{\text{clip}} - \eta\mathcal{H}. \tag{3}$$

The clipped loss compares rollouts within each scene group and limits the incentive for large changes in action likelihood. We use separate upper and lower clipping bounds. The entropy term $\mathcal{H}$, weighted by η, maintains exploration during manipulation and composition. Together, these choices help keep early updates conservative while successful behavior is still rare.

During skill learning, DAPO-style dynamic sampling filters uninformative groups and collects additional rollout groups within the configured collection budget [4]. With binary rewards, uniform outcomes provide no group-relative contrast. During dense acquisition, filtering instead checks within-group return variation, so partial progress can provide a learning signal before a group contains a successful skill execution.

During full-task continuation, the inherited VLA and LoRA remain frozen while the residual and exploration scale are trained with a fresh optimizer and the same milestone rewards. The controller adds Gaussian exploration noise and an episode-level offset to its predicted mean before clipping the command for execution. The offset is sampled once per episode and applied during selected phases. We store the original sample and the applied offset, then use both to evaluate its likelihood under the old and new policies [9].

Let $\mu_{\theta,t}$ denote the controller's predicted mean, σ_θ the learned exploration scale, and b_t the applied offset. The sampled command u_t and executed action a_t are

$$\begin{aligned} u_t &\sim \mathcal{N}(\mu_{\theta,t} + b_t, \operatorname{diag}(\sigma_\theta^2)), \\ a_t &= \operatorname{clip}(u_t, -1, 1). \end{aligned} \tag{4}$$

The likelihood ratio scores the stored u_t conditional on the same b_t. Section IV-C reports complete-task performance.

IV. Experiments

We evaluate skill retention and complete-task adaptation in CDPR simulation, and study intermediate feedback separately on LIBERO-Spatial.

The CDPR simulation uses MuJoCo Warp/MJ-Lab, four object categories (apple, orange, potato, and tomato), and plate or bowl destinations. The action interface controls Cartesian translation, yaw, and the gripper; a fixed inverse-kinematics controller converts Cartesian targets to cable lengths.

No human demonstrations are collected on the CDPR. Skill consolidation uses successful policy rollouts. Before full-task RL, the policy is additionally fitted to automatically generated complete-task trajectories from a scripted oracle and assisted rollouts of learned skills. Full-task RL and evaluation use unassisted empty-gripper starts.

A. Base-Policy Skill Learning and Retention

These measurements concern dense-reward training of simple CDPR skills and their consolidation into a base policy, before full-task composition. Each skill uses the instruction and reset protocol in Section III-A; placement tests start with the object already held. Learning a new skill can degrade previously learned behavior, although the same policy must retain all component skills to execute the complete multi-stage task.

In a forgetting evaluation, mean reaching success falls from 64.1% for the reaching reference to 5.3% after placement RL, over three evaluation rounds per condition. The recovery evaluation in Fig. 3 is a separate measurement: balanced self-imitation raises reaching success from 8.0% to 31.1%. Plate placement changes from 50.0% to 52.3%, and bowl placement from 23.05% to 27.65%. The bank combines earlier reaching

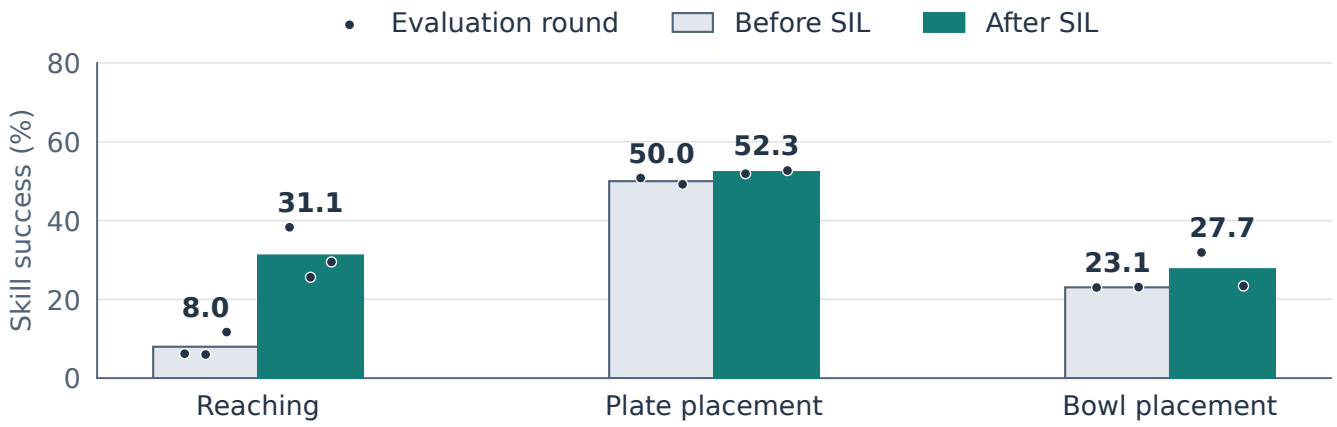


Fig. 3. Base-policy skill recovery before full-task composition. Bars show means and points show individual evaluation rounds: three for reaching and two for each placement skill, per condition. Reaching uses the instruction *move to <object>*; placement starts with the object already held. Reaching partially recovers while both placement means increase.

successes with current placement successes, balancing the retained data by instruction.

Recovery is partial, with placement performance retained while reaching improves. Table I summarizes skill-specific RL references and a shared policy after subsequent consolidation. Evaluation reset distributions differ across stages. The shared policy succeeds on all four skills, although pickup remains weak. These skill evaluations do not isolate the contribution of self-imitation to final full-task success.

TABLE I
SKILL SUCCESS (%) BY TRAINING STAGE. PLACEMENT STARTS WITH THE OBJECT HELD; — DENOTES AN UNREPORTED RESULT.

Training stage	Reach	Pickup	Plate	Bowl
Reaching RL	64.1	—	—	—
Pickup RL	—	26.0	—	—
Placement RL	—	—	71.0	32.0
After consolidation	47.8	11.9	73.8	53.5

B. Intermediate Feedback on LIBERO-Spatial

This experiment compares feedback for complete manipulation instructions, using OpenVLA-OFT and SimpleVLA-RL to hold the backbone and training framework fixed across reward conditions [2], [5]. It tests an ingredient of the adaptation pipeline; the complete training schedule is evaluated on the CDPR.

We fine-tune with LoRA on the ten LIBERO-Spatial tasks [10], using the feedback variants in Section III-C. All runs share a 61.0% initialization and use 300 validation rollouts per endpoint. The terminal-only baseline is trained with SimpleVLA-RL under the same 160-update budget and terminal verifier coefficient 5 as the dense-and-phase condition. The phase-only condition uses phase-completion bonuses with terminal verifier coefficient 1 and a 110-update budget. Dense-and-phase feedback reaches 65.3% success versus 61.7% for terminal-only SimpleVLA-RL at the same update budget (Table II). Phase-only feedback reaches 62.3% under its separate reward configuration and budget.

The comparison examines the effect of intermediate feedback on complete-task learning. In the CDPR pipeline, binary milestones provide this feedback during composition, while continuous progress rewards are used for initial skill acquisition. This choice limits continuous reward design to the simpler skills.

TABLE II
LIBERO-SPATIAL REWARD COMPARISON FROM A 61.0% INITIALIZATION. SUCCESS IS REPORTED IN PERCENT; Δ IS THE GAIN IN PERCENTAGE POINTS.

Configuration	Updates	Success	Δ
Terminal only (SimpleVLA-RL [5])	160	61.7	+0.7
Dense + phase	160	65.3	+4.3
Phase-only	110	62.3	+1.3

C. SmolVLA: Complete-Task Adaptation

Strict success requires grasp, held lift, intentional release, and correct placement, without carry slip or wrong-placement events.

Development evaluation uses 512 distinct scenes, each repeated four times, for 2,048 full-task attempts: 1,028 plate and 1,020 bowl attempts. The fixed 128-scene validation panel used during training is excluded. Each attempt starts with an empty gripper and uses one full-task instruction, without teacher assistance. Residual actions are deterministic while the VLA prior retains unit-scale sampling noise. All attempts, including numerical failures, remain in the denominator. These development scenes support model selection; the final-test split remains unused. The adapted controller completes 1,416 of 2,048 development attempts (69.14% strict success; Table III). Physical grasp occurs in 92.14% of attempts and held lift in 89.45%. Completion rates are 774/1,028 (75.29%) for plates and 642/1,020 (62.94%) for bowls. Carry slips (19.43%) and

wrong placements (14.89%) remain the main failure modes. A matched comparison with the preceding controller yields an 8.35 percentage-point strict-success gain (scene-clustered paired 95% interval: 6.05–10.69 points; Holm-adjusted $p = 0.00005$), with improved grasp and lift. Numerical failures are 24/2,048 (1.17%); their change is not distinguishable from zero in this comparison.

TABLE III
CDPR OFF-PANEL DEVELOPMENT EVALUATION: 512 SCENES REPEATED FOUR TIMES (2,048 ATTEMPTS).

Outcome	Attempts (%)
Physical grasp	92.14
Held lift	89.45
Strict completion (1,416/2,048)	69.14
Carry slip	19.43
Wrong placement	14.89

D. Additional Embodiment: WidowX-200 Simulation

A completed run on a simulated five-DoF WidowX-200 arm yields 33.30% reaching and 24.02% pickup success, with 1,024 attempts per skill. Placement from an already-held object succeeds in 72/520 bowl and 293/504 plate attempts (35.64% pooled). Empty-hand composition succeeds in 32/520 bowl and 41/504 plate attempts (7.13% pooled). These composed resets align the gripper above the object and omit a distant approach, so the rates are not directly comparable with CDPR full-task success. Earlier skill losses persist through consolidation; composition SFT lowers imitation error without improving rollout success. This initial port exposes unresolved retention and grasp-to-placement transitions.

E. Scope of the Evidence

The LIBERO comparison concerns reward configurations, whereas the CDPR experiment evaluates the full adaptation procedure. The phase-only LIBERO condition differs in reward weights and training budget, so it does not isolate the effect of removing continuous progress feedback. The CDPR result comes from one training lineage, evaluated on development scenes outside its fixed training-validation panel. The four rollout repeats are not independent training seeds, and the final-test split is unused. Retention measurements cover a small number of rounds. The preliminary WidowX-200 result does not establish reliable adaptation across embodiments. Matched stage ablations are needed to quantify the contribution of each part of the pipeline.

V. CONCLUSION

We presented a staged pipeline for adapting a pretrained VLA to a target robot without collecting human demonstrations on that embodiment. Adaptation begins with dense-reward RL of a new residual controller, followed by binary refinement, successful-trajectory consolidation, and milestone-based full-task learning. Skill evaluations show partial recovery of earlier behavior after balanced self-imitation, and the adapted CDPR controller reaches 69.14% strict success in off-panel simulation development evaluation. The initial WidowX-200 port reveals remaining retention and composition limitations. The complementary LIBERO-Spatial experiment compares intermediate and terminal feedback under a common training setup.

VI. FUTURE WORK

Further WidowX-200 work will verify sustained grasp stability, reconcile teacher performance across reset distributions, and strengthen per-skill retention before complete-task training. Additional experiments will compare training stages under matched budgets, repeat training across independent seeds, and evaluate the reserved final-test scenes. Automatic phase detection and progress estimation could also reduce the need for manually designed reward predicates.

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